Chapter 19

Chemical Coordination and Integration

The neural system provides a point-to-point rapid coordination among organs, but neural coordination is fast and short-lived.

Why a second system is needed

- the nerve fibres do not innervate all the cells of the body, and cellular functions need to be regulated continuously
- so a special kind of coordination and integration has to be provided, and this function is carried out by hormones
- the neural system and the endocrine system jointly coordinate and regulate the physiological functions in the body

Endocrine glands and their secretions

- endocrine glands lack ducts and are hence called ductless glands, and their secretions are called hormones
- the classical definition of a hormone was a chemical produced by an endocrine gland, released into the blood and transported to a distantly located target organ
- the current scientific definition is that hormones are non-nutrient chemicals which act as intercellular messengers and are produced in trace amounts
- the new definition covers a number of new molecules besides the hormones secreted by the organised endocrine glands

Across the animal kingdom

- invertebrates possess very simple endocrine systems with few hormones
- in the vertebrates a large number of chemicals act as hormones and provide coordination
- the human endocrine system is the one described in this chapter

In this chapter

19.1 The human endocrine system

19.3 The pituitary gland

19.6 The thyroid gland

19.10 The adrenal gland

19.12 The pancreas

19.17 Mechanism of hormone action

Hypothalamus and pineal, within the brain

Pituitary, just below the hypothalamus

Thyroid and parathyroid in the neck, thymus behind the sternum

Pancreas and the paired adrenals in the abdomen

Ovary in the female, testis in the male

Figure 19.A Location of endocrine glands

19.1 The human endocrine system

What makes up the system

- the endocrine glands and the hormone producing diffused tissues and cells located in different parts of our body constitute the endocrine system
- the pituitary, pineal, thyroid, adrenal, pancreas, parathyroid, thymus and gonads, that is testis in males and ovary in females, are the organised endocrine bodies in our body
- some other organs, such as the gastrointestinal tract, the liver, the kidney and the heart, also produce hormones

19.2 The hypothalamus

Releasing and inhibiting hormones

- the hypothalamus is the basal part of the diencephalon in the forebrain, and it regulates a wide spectrum of body functions
- it contains several groups of neurosecretory cells called nuclei, which produce hormones that regulate the synthesis and secretion of the pituitary hormones
- these hormones are of two types, the releasing hormones, which stimulate the secretion of pituitary hormones, and the inhibiting hormones, which inhibit it
- gonadotrophin releasing hormone (GnRH) stimulates the pituitary synthesis and release of gonadotrophins, whereas somatostatin from the hypothalamus inhibits the release of growth hormone from the pituitary

How they reach the pituitary

- these hormones originate in the hypothalamic neurons, pass through axons and are released from their nerve endings
- they reach the pituitary gland through a portal circulatory system and regulate the functions of the anterior pituitary
- the posterior pituitary is under the direct neural regulation of the hypothalamus

Hypothalamus above, with its hypothalamic neurons

Portal circulation, carrying hormones down to the gland

Anterior pituitary, regulated through that portal blood

Posterior pituitary, under direct neural control

Figure 19.B Diagrammatic representation of pituitary and its relationship with hypothalamus

19.3 The pituitary gland

Its parts and their hormones

- the pituitary gland lies in a bony cavity called the sella tursica and is attached to the hypothalamus by a stalk
- anatomically it divides into an adenohypophysis and a neurohypophysis, the adenohypophysis consisting of two portions, the pars distalis and the pars intermedia
- the pars distalis, commonly called the anterior pituitary, produces growth hormone (GH), prolactin (PRL), thyroid stimulating hormone (TSH), adrenocorticotrophic hormone (ACTH), luteinizing hormone (LH) and follicle stimulating hormone (FSH), while the pars intermedia secretes only melanocyte stimulating hormone (MSH), although in humans it is almost merged with the pars distalis
- the neurohypophysis or pars nervosa, also known as the posterior pituitary, stores and releases oxytocin and vasopressin, which are actually synthesised by the hypothalamus and transported axonally to it

19.4 What the pituitary hormones do

Growth hormone and prolactin

- over-secretion of GH stimulates abnormal growth of the body, leading to gigantism, and low secretion of GH results in stunted growth, that is pituitary dwarfism
- excess secretion of growth hormone in adults, especially in middle age, can result in severe disfigurement, especially of the face, called acromegaly, which may lead to serious complications and premature death if unchecked
- the disease is hard to diagnose in the early stages and often goes undetected for many years, until changes in external features become noticeable
- prolactin regulates the growth of the mammary glands and the formation of milk in them

The trophic hormones

- TSH stimulates the synthesis and secretion of thyroid hormones from the thyroid gland
- ACTH stimulates the synthesis and secretion of the steroid hormones called glucocorticoids from the adrenal cortex
- LH and FSH stimulate gonadal activity and are hence called gonadotrophins, LH stimulating in males the synthesis and secretion of the hormones called androgens from the testis, with FSH and androgens together regulating spermatogenesis
- in females LH induces ovulation of fully mature follicles, the graafian follicles, and maintains the corpus luteum formed from their remnants after ovulation, while FSH stimulates the growth and development of the ovarian follicles

Thyroid, two lobes on either side of the trachea

Isthmus, the thin flap joining the lobes, seen from the front

Parathyroids, four glands on the back of the thyroid

One pair of them in each lobe, seen from behind

Figure 19.C Diagrammatic view of the position of Thyroid and Parathyroid (a) Ventral side (b) Dorsal side

19.5 Oxytocin, vasopressin and the pineal gland

The remaining pituitary hormones

- MSH acts on the melanocytes, the melanin containing cells, and regulates the pigmentation of the skin
- oxytocin acts on the smooth muscles of our body and stimulates their contraction, and in females it stimulates a vigorous contraction of the uterus at the time of child birth and milk ejection from the mammary gland
- vasopressin acts mainly at the kidney and stimulates the resorption of water and electrolytes by the distal tubules, thereby reducing the loss of water through urine, that is diuresis, which is why it is also called antidiuretic hormone (ADH)
- an impairment affecting the synthesis or release of ADH leaves the kidney less able to conserve water, leading to water loss and dehydration, a condition known as diabetes insipidus

The pineal gland

- the pineal gland is located on the dorsal side of the forebrain and secretes a hormone called melatonin
- melatonin matters greatly in the regulation of the 24-hour, that is diurnal, rhythm of our body, helping maintain the normal rhythms of the sleep-wake cycle and of body temperature
- it also influences metabolism, pigmentation, the menstrual cycle and our defense capability

19.6 The thyroid gland

Structure and hormones

- the thyroid gland is composed of two lobes located on either side of the trachea, interconnected by a thin flap of connective tissue called the isthmus
- it is composed of follicles and stromal tissues, each thyroid follicle being composed of follicular cells enclosing a cavity
- these follicular cells synthesise two hormones, tetraiodothyronine or thyroxine (T4) and triiodothyronine (T3), and iodine is essential for the normal rate of hormone synthesis in the thyroid
- deficiency of iodine in our diet results in hypothyroidism and enlargement of the thyroid gland, commonly called goitre, and hypothyroidism during pregnancy causes defective development and maturation of the growing baby, leading to stunted growth, that is cretinism

19.7 Thyroid disorders and functions

Too little and too much

- cretinism also brings mental retardation, low intelligence quotient, abnormal skin, deaf-mutism and the like, and in adult women hypothyroidism may make the menstrual cycle irregular
- cancer of the thyroid gland, or the development of nodules on it, raises the rate of synthesis and secretion of the thyroid hormones to abnormally high levels, a condition called hyperthyroidism, which adversely affects the body physiology
- exopthalmic goitre is a form of hyperthyroidism, marked by enlargement of the thyroid gland, protrusion of the eyeballs, increased basal metabolic rate and weight loss, and is also called Graves' disease

What the thyroid hormones do

- they matter greatly in the regulation of the basal metabolic rate
- they support the process of red blood cell formation
- they control the metabolism of carbohydrates, proteins and fats, and they influence the maintenance of water and electrolyte balance
- the thyroid gland also secretes a protein hormone called thyrocalcitonin (TCT), which regulates the blood calcium levels

19.8 The parathyroid gland

Parathyroid hormone

- in humans four parathyroid glands are present on the back side of the thyroid gland, one pair in each of its two lobes
- they secrete a peptide hormone called parathyroid hormone (PTH), whose secretion is regulated by the circulating levels of calcium ions
- PTH increases the Ca^{2+} levels in the blood, acting on bones and stimulating the process of bone resorption, that is dissolution or demineralisation
- it also stimulates the reabsorption of Ca^{2+} by the renal tubules and increases Ca^{2+} absorption from the digested food, so it is a hypercalcemic hormone, and along with TCT it matters greatly in calcium balance in the body

19.9 The thymus

Thymosins and immunity

- the thymus gland is a lobular structure located between the lungs, behind the sternum on the ventral side of the aorta, and it matters greatly in the development of the immune system
- this gland secretes peptide hormones called thymosins, which matter greatly in the differentiation of T-lymphocytes, which provide cell-mediated immunity
- thymosins also promote the production of antibodies to provide humoral immunity
- the thymus is degenerated in old individuals, so the production of thymosins falls and the immune responses of old persons become weak

Adrenal gland, sitting above the kidney

Adrenal medulla, the centrally located tissue

Adrenal cortex, the tissue lying outside it

Kidney below, in the same view

Figure 19.D Diagrammatic representation of : (a) Adrenal gland above kidney (b) Section showing two parts of adrenal gland

19.10 The adrenal gland

Two tissues, and one deficiency

- our body has one pair of adrenal glands, one above each kidney
- the gland is composed of two types of tissue, the centrally located adrenal medulla and, outside it, the adrenal cortex
- underproduction of hormones by the adrenal cortex alters carbohydrate metabolism, causing acute weakness and fatigue, and leading to a disease called Addison's disease

The catecholamines

- the adrenal medulla secretes two hormones, adrenaline or epinephrine and noradrenaline or norepinephrine, commonly called the catecholamines
- they are rapidly secreted in response to stress of any kind and during emergency situations, and are called the emergency hormones or the hormones of fight or flight
- they increase alertness, pupilary dilation, piloerection, that is the raising of hairs, and sweating, and both increase the heart beat, the strength of heart contraction and the rate of respiration
- catecholamines also stimulate the breakdown of glycogen

The rest of what catecholamines do

- the breakdown of glycogen raises the concentration of glucose in blood
- they also stimulate the breakdown of lipids and proteins

The layers of the cortex

- the adrenal cortex can be divided into three layers, the zona reticularis as the inner layer, the zona fasciculata as the middle layer and the zona glomerulosa as the outer layer
- the cortex secretes many hormones, commonly called corticoids
- the corticoids involved in carbohydrate metabolism are the glucocorticoids, cortisol being the main one in our body, and those regulating the balance of water and electrolytes are the mineralocorticoids, aldosterone being the main one

19.11 The adrenal cortex

What the corticoids do

- glucocorticoids stimulate gluconeogenesis, lipolysis and proteolysis, and inhibit the cellular uptake and utilisation of amino acids
- cortisol is also involved in maintaining the cardio-vascular system as well as kidney function, it produces anti-inflammatory reactions and suppresses the immune response, and it stimulates RBC production
- aldosterone acts mainly at the renal tubules and stimulates the reabsorption of Na+ and water and the excretion of K+ and phosphate ions, so it helps maintain electrolytes, body fluid volume, osmotic pressure and blood pressure
- small amounts of androgenic steroids are also secreted by the adrenal cortex, and these play a role in the growth of axial hair, pubic hair and facial hair during puberty

19.12 The pancreas

Islets of Langerhans, and glucagon

- the pancreas is a composite gland which acts as both an exocrine and an endocrine gland, the endocrine pancreas consisting of the Islets of Langerhans
- there are about 1 to 2 million Islets of Langerhans in a normal human pancreas, representing only 1 to 2 per cent of the pancreatic tissue, and the two main types of cell in an islet are the α-cells, which secrete glucagon, and the β-cells, which secrete insulin
- glucagon is a peptide hormone that matters greatly in maintaining the normal blood glucose levels, acting mainly on the liver cells or hepatocytes and stimulating glycogenolysis, which raises blood sugar, that is hyperglycemia
- it also stimulates gluconeogenesis, which adds to the hyperglycemia, and it reduces cellular glucose uptake and utilisation, so glucagon is a hyperglycemic hormone

19.13 Insulin and diabetes mellitus

Insulin

- insulin is a peptide hormone which matters greatly in the regulation of glucose homeostasis
- it acts mainly on hepatocytes and adipocytes, the cells of adipose tissue, and enhances cellular glucose uptake and utilisation, so glucose moves rapidly from blood into these cells and the blood glucose level falls, that is hypoglycemia
- insulin also stimulates the conversion of glucose to glycogen, that is glycogenesis, in the target cells, and glucose homeostasis in blood is thus maintained jointly by insulin and glucagons
- prolonged hyperglycemia leads to a complex disorder called diabetes mellitus, associated with loss of glucose through urine and the formation of harmful compounds known as ketone bodies, and diabetic patients are successfully treated with insulin therapy

19.14 The testis

Structure and androgens

- a pair of testis is present in the scrotal sac, outside the abdomen, of male individuals, and the testis performs dual functions, as a primary sex organ and as an endocrine gland
- it is composed of seminiferous tubules and stromal or interstitial tissue, and the Leydig or interstitial cells present in the intertubular spaces produce a group of hormones called androgens, mainly testosterone
- androgens regulate the development, maturation and functions of the male accessory sex organs such as the epididymis, vas deferens, seminal vesicles, prostate gland and urethra, and they stimulate muscular growth, the growth of facial and axillary hair, aggressiveness and a low pitch of voice
- they play a major stimulatory role in spermatogenesis, the formation of spermatozoa, they act on the central neural system and influence male sexual behaviour or libido, and they produce anabolic, that is synthetic, effects on protein and carbohydrate metabolism

19.15 The ovary

Estrogen and progesterone

- females have a pair of ovaries located in the abdomen, the ovary being the primary female sex organ, which produces one ovum during each menstrual cycle
- it also produces two groups of steroid hormones, estrogen and progesterone
- the ovary is composed of ovarian follicles and stromal tissues, the estrogen being synthesised and secreted mainly by the growing ovarian follicles
- after ovulation the ruptured follicle is converted to a structure called the corpus luteum, which secretes mainly progesterone

What the ovarian hormones do

- estrogens produce wide ranging actions such as the stimulation of growth and activities of the female secondary sex organs, the development of growing ovarian follicles, the appearance of female secondary sex characters such as a high pitch of voice, and mammary gland development, and they also regulate female sexual behaviour
- progesterone supports pregnancy
- progesterone also acts on the mammary glands and stimulates the formation of alveoli, the sac-like structures which store milk, and milk secretion

19.16 Hormones of the heart, kidney and gut

Heart and kidney

- hormones are also secreted by some tissues which are not endocrine glands
- the atrial wall of our heart secretes a very important peptide hormone called atrial natriuretic factor (ANF), which decreases blood pressure, being secreted when blood pressure rises and causing dilation of the blood vessels, which brings the pressure down
- the juxtaglomerular cells of the kidney produce a peptide hormone called erythropoietin, which stimulates erythropoiesis, the formation of RBC

The gastrointestinal hormones

- endocrine cells in different parts of the gastro-intestinal tract secrete four major peptide hormones, gastrin, secretin, cholecystokinin (CCK) and gastric inhibitory peptide (GIP)
- gastrin acts on the gastric glands and stimulates the secretion of hydrochloric acid and pepsinogen, while secretin acts on the exocrine pancreas and stimulates the secretion of water and bicarbonate ions
- CCK acts on both the pancreas and the gall bladder and stimulates the secretion of pancreatic enzymes and bile juice respectively, while GIP inhibits gastric secretion and motility
- several other non-endocrine tissues secrete hormones called growth factors, which are essential for the normal growth of tissues and for their repair and regeneration

19.17 Mechanism of hormone action

Hormone receptors

- hormones produce their effects on target tissues by binding to specific proteins called hormone receptors, located in the target tissues only
- receptors present on the cell membrane of the target cells are called membrane-bound receptors, and those present inside the target cell are called intracellular receptors, mostly nuclear receptors present in the nucleus
- binding of a hormone to its receptor leads to the formation of a hormone-receptor complex, and each receptor is specific to one hormone only
- formation of the complex leads to certain biochemical changes in the target tissue, so that target tissue metabolism, and hence physiological function, is regulated by hormones

19.18 Classes of hormone and second messengers

Four chemical groups

- on the basis of their chemical nature, hormones can be divided into groups
- peptide, polypeptide and protein hormones, such as insulin, glucagon, the pituitary hormones and the hypothalamic hormones
- steroids, such as cortisol, testosterone, estradiol and progesterone
- iodothyronines, that is the thyroid hormones, and amino-acid derivatives, such as epinephrine

Two routes into the cell

- hormones which interact with membrane-bound receptors normally do not enter the target cell, but generate second messengers such as cyclic AMP, IP3 and Ca++, which in turn regulate cellular metabolism
- hormones which interact with intracellular receptors, such as the steroid hormones and the iodothyronines, mostly regulate gene expression or chromosome function by the interaction of the hormone-receptor complex with the genome
- cumulative biochemical actions result in physiological and developmental effects

Protein hormone, binding a receptor on the cell membrane

Second messenger, generated inside the cell

Steroid hormone, passing in to an intracellular receptor

Hormone-receptor complex, acting on the genome

Figure 19.E Diagramatic representation of the mechanism of hormone action : (a) Protein hormone (b) Steroid hormone

19.19 Summary

The chemical coordinators

- there are special chemicals which act as hormones and provide chemical coordination, integration and regulation in the human body
- these hormones regulate metabolism and the growth and development of our organs, the endocrine glands or certain cells
- the endocrine system is composed of the hypothalamus, pituitary and pineal, the thyroid, adrenal, pancreas, parathyroid and thymus, and the gonads, that is testis and ovary, and some other organs such as the gastrointestinal tract, kidney and heart also produce hormones

Pituitary and pineal

- the pituitary gland is divided into three major parts, the pars distalis, the pars intermedia and the pars nervosa
- the pars distalis produces six trophic hormones, the pars intermedia secretes only one, and the pars nervosa or neurohypophysis secretes two
- the pituitary hormones regulate the growth and development of somatic tissues and the activities of the peripheral endocrine glands
- the pineal gland secretes melatonin, which matters greatly in the regulation of the 24-hour or diurnal rhythms of our body, such as the rhythms of sleep and the state of being awake and of body temperature

19.20 Summary, continued

Thyroid, parathyroid, thymus

- the thyroid gland hormones matter greatly in the regulation of the basal metabolic rate, the development and maturation of the central neural system, erythropoiesis, the metabolism of carbohydrates, proteins and fats, and the menstrual cycle
- another thyroid hormone, thyrocalcitonin, regulates calcium levels in our blood by decreasing them
- the parathyroid glands secrete parathyroid hormone (PTH), which increases the blood Ca2+ levels and plays a major role in calcium homeostasis
- the thymus gland secretes thymosins, which play a major role in the differentiation of T-lymphocytes, which provide cell-mediated immunity, and thymosins also increase the production of antibodies to provide humoral immunity

Adrenal and pancreas

- the adrenal gland is composed of the centrally located adrenal medulla and the outer adrenal cortex, the medulla secreting epinephrine and norepinephrine, which increase alertness, pupilary dilation, piloerection, sweating, heart beat, strength of heart contraction, rate of respiration, glycogenolysis, lipolysis and proteolysis
- the adrenal cortex secretes glucocorticoids and mineralocorticoids, the glucocorticoids stimulating gluconeogenesis, lipolysis, proteolysis, erythropoiesis, the cardio-vascular system, blood pressure and the glomerular filtration rate, and inhibiting inflammatory reactions by suppressing the immune response, while the mineralocorticoids regulate the water and electrolyte contents of the body
- the endocrine pancreas secretes glucagon and insulin, glucagon stimulating glycogenolysis and gluconeogenesis, which results in hyperglycemia, and insulin stimulating cellular glucose uptake and utilisation and glycogenesis, which results in hypoglycemia
- insulin deficiency or insulin resistance, or both, result in a disease called diabetes mellitus

Gonads, heart, kidney and gut

- the testis secretes androgens, which stimulate the development, maturation and functions of the male accessory sex organs, the appearance of the male secondary sex characters, spermatogenesis, male sexual behaviour, anabolic pathways and erythropoiesis
- the ovary secretes estrogen and progesterone, estrogen stimulating the growth and development of the female accessory sex organs and secondary sex characters, and progesterone playing a major role in the maintenance of pregnancy as well as in mammary gland development and lactation
- the atrial wall of the heart produces atrial natriuretic factor, which decreases the blood pressure, and the kidney produces erythropoietin, which stimulates erythropoiesis
- the gastrointestinal tract secretes gastrin, secretin, cholecystokinin and gastric inhibitory peptide, and these hormones regulate the secretion of digestive juices and help in digestion

19.21 Exercises

From this chapter

1. Define the following: (a) Exocrine gland (b) Endocrine gland (c) Hormone
2. Diagrammatically indicate the location of the various endocrine glands in our body.
3. List the hormones secreted by the following: (a) Hypothalamus (b) Pituitary (c) Thyroid (d) Parathyroid (e) Adrenal (f) Pancreas (g) Testis (h) Ovary (i) Thymus (j) Atrium (k) Kidney (l) G-I Tract
4. Fill in the blanks: Hormones, Target gland. (a) Hypothalamic hormones _____ (b) Thyrotrophin (TSH) _____ (c) Corticotrophin (ACTH) _____ (d) Gonadotrophins (LH, FSH) _____ (e) Melanotrophin (MSH) _____
5. Write short notes on the functions of the following hormones: (a) Parathyroid hormone (PTH) (b) Thyroid hormones (c) Thymosins (d) Androgens (e) Estrogens (f) Insulin and Glucagon
6. Give example(s) of: (a) Hyperglycemic hormone and hypoglycemic hormone (b) Hypercalcemic hormone (c) Gonadotrophic hormones (d) Progestational hormone (e) Blood pressure lowering hormone (f) Androgens and estrogens
7. Which hormonal deficiency is responsible for the following: (a) Diabetes mellitus (b) Goitre (c) Cretinism
8. Briefly mention the mechanism of action of FSH.
9. Match the following: Column I (a) T4 (b) PTH (c) GnRH (d) LH , Column II (i) Hypothalamus (ii) Thyroid (iii) Pituitary (iv) Parathyroid

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